

Skills Summary

One-Step Equations with Rational Coefficients

Use this reference while completing your worksheet or when studying.

Skill 1 — Undo adding or subtracting

Rule: an equation is a balance — do the same thing to *both* sides. To undo $x + a = b$, subtract a from both sides; to undo $x - a = b$, add a .

Example: Solve $x + 6 = 13$

Step 1. The x has 6 added to it, so undo the addition by subtracting 6 from both sides of the balance.

Step 2. $x = 13 - 6 = 7$. Check: $7 + 6 = 13$. ✓

$$\Rightarrow x = 7$$

Try it: Solve $x - 5 = 9$

check: $14 = x$

Skill 2 — Divide both sides

Rule: to undo $ax = b$, divide both sides by a : $x = b/a$. A fraction answer in simplest form is a perfectly good answer.

Example: Solve $4x = 30$

Step 1. The x is multiplied by 4, so undo it by dividing both sides by 4, then simplify the fraction.

Step 2. $x = \frac{30}{4} = \frac{15}{2}$ after dividing top and bottom by 2.

$$\Rightarrow x = \frac{15}{2}$$

Try it: Solve $8x = 20$

check: $\frac{5}{2} = x$

Skill 3 — Multiply by the reciprocal

Rule: to undo a *fraction* coefficient $\frac{a}{b}x = c$ in one step, multiply both sides by the reciprocal $\frac{b}{a}$, because $\frac{b}{a} \cdot \frac{a}{b} = 1$.

Example: Solve $\frac{3}{5}x = 12$

Step 1. The coefficient is the fraction three-fifths, so multiply both sides by its reciprocal five-thirds to leave x alone.

Step 2. $x = 12 \cdot \frac{5}{3} = \frac{60}{3} = 20$.

$$\Rightarrow x = 20$$

Try it: Solve $\frac{2}{7}x = 6$

check: $21 = x$

(!) Watch out: dividing by a fraction and multiplying by its reciprocal are the same move — never divide by only the numerator.

Skill 4 — Undo a division

Rule: to undo $\frac{x}{a} = b$, multiply both sides by a : $x = ab$. Picture a tape cut into a equal parts, each worth b .

Example: Solve $\frac{x}{6} = 7$

Step 1. The x is divided by 6, so undo the division by multiplying both sides by 6.

Step 2. $x = 7 \cdot 6 = 42$. Check: $42 \div 6 = 7$. ✓

$$\Rightarrow x = 42$$

Try it: Solve $\frac{x}{4} = 9$

check: $36 = x$